

CS & IT ENGINEERING



Computer Network

IPv4 Address

DPP - 01 Discussion Notes



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[MCQ]



#Q. Consider a class-less IPv4 network where network id bits assigned to this network are "01010011 10011011 0110", identify correct decimal representation of network address which is used to represent this network?

Network Address : 83.155.96.0

01010011 10011011 01100000 00000000
20 bit NetID 12 bit HostID

A 83.155.0.0

☒ **B** 83.155.96.0

C 83.155.96.255

D 83.155.111.255

Ans: B

[MCQ]



#Q. Consider a class-less IPv4 network where network id bits assigned to this network are "01010011 10011011 0110", identify correct decimal representation of broadcast address which is used to broadcast a packet in this network?

Broadcast Address: 83.155.111.255

01010011 10011011 01101111 11111111
20 bit Net ID 12 bit Host ID

- A** 83.155.0.0
- B** 83.155.96.0
- C** 83.155.96.255
- D** 83.155.111.255

Ans: D

[MCQ]



#Q. Consider a class-less IPv4 network, the network address of this network is "205 . 215 . 104 . 0" and size of network identifier field is 22 bits, which one of the following is a broadcast address of this network?

A 205 . 215 . 104 . 0

B 205 . 215 . 104 . 255

☒ **C** 205 . 215 . 107 . 255

D 205 . 215 . 111 . 255

Ans: C

Network Address = 205.215.104.0 /22

205.215.01101000.00000000

NetID 22 bit

HostID 10 bit

Broadcast Address = 205.215.107.255

205.215.01101011.11111111

22 bit NetID

10 bit HostID

[MCQ]



#Q. Consider a class-less IPv4 network, the broadcast address of this network is "25 . 15 . 95 . 255" and size of network identifier field is 22 bits, which one of the following is a network address of this network ?

A 25 . 15 . 64 . 0

B 25 . 15 . 80 . 0

☒ **C** 25 . 15 . 92 . 0

D 25 . 15 . 95 . 0

Broadcast Address = 25 . 15 . 95 . 255 /22

25 . 15 . 01011111 . 11111111

NetID 22 bit HostID 10 bit

Network Address = 25 . 15 . 92 . 0

25 . 15 . 01011100 . 00000000

NetID 22 bit HostID (10 bit)

Ans: C

[MSQ]



#Q. Consider a class-less IPv4 network, the network address of this network is "140.190.208.0", which is/are can be a broadcast address of this network?

☒ A

140.190.208.63

☐ B

140.190.223.0

☐ C

140.190.223.63

☒ D

140.190.223.255

Network Address = 140.190.208.0

140.190.11010000.00000000
Min^m 20 bit NetID Max^m 12 bit HostID

A) 140.190.208.63

140.190.11010000.00111111
26 bit NetID 6 bit HostID

Ans: A & D

B) 140.190.223.0
140.190.11011111.00000000

C) 140.190.223.63
140.190.11011111.00111111

6 bit Host ID

D) 140.190.223.255
140.190.11011111.11111111

20 bit NetID 12 bit Host ID

[MSQ]

#Q. Consider a class-less IPv4 network, the broadcast address of this network is "120 . 120 . 159 . 255", which is/are can be a network address of this network?

Broadcast Address = 120.120.159.255

120.120.10011111.11111111
 Min^m 19 bit Net ID Max^m 13 bit Host ID

A) 120.120.136.0
 120.120.10001000.00000000
 11 bit Host ID

- ☒ A 120.120.136.0
- ☒ B 120.120.144.0
- ☒ C 120.120.159.0
- ☒ D 120.120.159.192

Ans: B, C, & D

B) 120.120.144.0

120.120.100|0000.00000000

20 bit NetID

12 bit HostID

C) 120.120.159.0

120.120.100|1111.00000000

24 bit NetID

8 bit HostID

D) 120.120.159.192

120.120.100|1111.11000000

26 bit NetID

6 bit HostID

[MCQ]



#Q. Consider an IPv4 address of a class-less IPv4 network is "15 . 20 . 25 . 30 / 21", identify correct pair of first and last IP address of this network which can be assigned to hosts ?

A 15 . 20 . 24 . 0 and 15 . 20 . 31 . 255

B 15 . 20 . 24 . 0 and 15 . 20 . 31 . 254

C 15 . 20 . 24 . 1 and 15 . 20 . 31 . 255

D 15 . 20 . 24 . 1 and 15 . 20 . 31 . 254

IP Add. $\rightarrow 15.20.25.30/21$

$15.20.00011001.00011110/21$
21 bit NetID 11 bit HostID

First Host IP $\rightarrow 15.20.24.1/21$

$15.20.00011000.000000001/21$

Last Host IP $\rightarrow 15.20.31.254/21$

$15.20.00011111.11111110/21$

Ans: D

[NAT]



#Q. Consider a class-less IPv4 network, the number of bits in the network id field are 26, how many maximum number of hosts can be in this network?

Net ID $\rightarrow$ 26 bits

Host ID $\rightarrow (32 - 26) = 6$ bits

Max^m no. of hosts can be in this network

$$= (2^6 - 2) \text{ hosts}$$

$$= 62 \text{ hosts}$$

Ans = 62

[NAT]



#Q. Consider a class-less IPv4 network, the network mask of this network is "255 . 255 . 254 . 0", how many maximum number of hosts can be in this network?

Net mask = 255.255.254.0

255.255.11111110.00000000
NetID (23 bit) HostID 9 bit

Max^m no. of hosts can be in this network

$$\begin{aligned} &= (2^9 - 2) \text{ hosts} \\ &= 510 \text{ hosts} \end{aligned}$$

Ans = 510

[MSQ]



#Q. Consider network mask of a class-less IPv4 network is "255 . 255 . 240 . 0", identify which pairs of IP addresses could belong to this network?

☒ **A** 210 . 130 . 148 . 12 and 210 . 130 . 168 . 30 $240 \rightarrow 11110000$

☒ **B** 210 . 130 . 148 . 14 and 210 . 130 . 158 . 32 $148 \rightarrow 10010100$

☒ **C** 210 . 130 . 158 . 16 and 210 . 130 . 168 . 34 $168 \rightarrow 10101000$

☒ **D** 210 . 130 . 164 . 18 and 210 . 130 . 168 . 38 $158 \rightarrow 10011110$

$164 \rightarrow 10100100$

Ans: B & D

[MSQ]

#Q. Consider two computers A and B and their IP addresses are "110 . 120 . 159 . 125" and "110 . 120 . 149 . 191" respectively. Which network mask they can use if they belongs to same network ?

☒ A

255 . 255 . 224 . 0

$IP_A : 110.120.10011111.125$

☒ B

255 . 255 . 240 . 0

$IP_B : 110.120.10010101.191$

☐ C

255 . 255 . 248 . 0

Max^m 20 bit in netmask.

☐ D

255 . 255 . 252 . 0

Ans: A & B

[NAT]



#Q. Consider a class B IPv4 network with 6 bit subnetting, how many maximum number of hosts can be in each sub-network?

Class B:-

NetID $\Rightarrow$ 16 bits

SubnetID $\Rightarrow$ 6 bits

HostID $\Rightarrow (32 - 16 - 6) = 10$ bit

Max^m no. of hosts can be in each subnet

$$= (2^{10} - 2) \text{ hosts}$$

$$= 1022 \text{ hosts}$$

Ans = 1022

[MCQ]

#Q. Consider a class C IPv4 network with 4 bit subnetting, identify correct subnet (sub-network) mask?

A 255.255.255.224

B 255.255.255.240

C 255.255.255.248

D 255.255.255.252

Subnet Mask:-

255.255.255.11110000
 NetID 4 bit SubnetID 4 bit HostID

255.255.255.240

Ans: B

[MCQ]



#Q. Consider a class B IPv4 network and wishes to form subnets for 128 departments, identify correct subnet (sub-network) mask?

A 255.255.128.0

B 255.255.192.0

☒ **C** 255.255.254.0

D 255.255.255.0

Class B:-

Net ID $\Rightarrow$ 16 bits

No. of subnets = 128 department

Subnet ID $\Rightarrow \lceil \log_2(128) \rceil$ bits = 7 bits

Subnet Mask: 255.255.254.0

255.255. 11111110 . 00000000
Net ID 7 bit Subnet ID 9 bit Host ID

Ans: C

[MSQ]



#Q. Consider a class C network with four bit subnetting, the network address of the network is "203 . 204 . 205 . 0". Which is/are can be a valid sub-network (subnet) address of a sub-network belongs to this network ?

☒ A 203 . 204 . 205 . 8

☒ B 203 . 204 . 205 . 16

☒ C 203 . 204 . 205 . 24

☒ D 203 . 204 . 205 . 32

Subnet Address:

----- 0000
Subnet Host

8 → 0000 | 000

16 → 000 | 0000

24 → 000 | 1000

32 → 0010 | 0000

Ans: B & D

[MSQ]



#Q. Consider a class B network with three bit subnetting, the network address of the network is "150 . 140 . 0 . 0". Which is/are can be a valid pair of subnet address and subnet broadcast address of a sub-network (subnet) ?

☒ A 150 . 140 . 64 . 0 and 150 . 140 . 127 . 255

☒ B 150 . 140 . 96 . 0 and 150 . 140 . 127 . 255

☒ C 150 . 140 . 128 . 0 and 150 . 140 . 191 . 255

☒ D 150 . 140 . 160 . 0 and 150 . 140 . 191 . 255

Ans. B & D

A) 64 → 010 00000
127 → 011 11111

B) 96 → 011 00000
127 → 011 11111

C) 128 → 100 00000
191 → 101 11111

D) 160 → 101 00000
191 → 101 11111

Subnet Add. = 150.140. — — — 00000.000000000



Broadcast Add. = 150.140. — — — 11111.111111111

[MSQ]

#Q. Consider a subnetted class C network, the broadcast address of a subnet is "220.210.210.31", which is/are can be a subnet mask of this subnet?

Class C:-

Subnet Broadcast Add. = 220.210.210.31

220.210.210.00011111
NetID Min^m Max^m 5 bit
 3 bit Host
 Subnetting

Subnet Mask can have 27 to 30 one's

☒ A 255.255.255.128

☒ B 255.255.255.192

☒ C 255.255.255.224

☒ D 255.255.255.240

Ans: C & D

[NAT]



#Q. Consider a class-less IPv4 network with four bit subnetting, how many maximum number of hosts will be reduced in the network after subnetting? [Note : Subnet all zero and all one are allowed]

* 4 bit subnetting

Reduction in no. of hosts after subnetting

$$= \left[(2^4) * 2 - 2 \right] = 30 \text{ hosts}$$

Ans = 30

#Q. Consider an Internet Service Provider (ISP) with an available IP addresses space is "25 . 28 . 144 . 0 / 21". The ISP wants to give half of this chunk of addresses to Organization A, and a quarter to Organization B, while retaining the remaining with itself. Which of the following is a valid allocation of addresses to A and B ?

☒ **A** 25 . 28 . 144 . 0 / 22 and 25 . 28 . 148 . 0 / 23

☒ **B** 25 . 28 . 144 . 0 / 22 and 25 . 28 . 150 . 0 / 23

☒ **C** 25 . 28 . 148 . 0 / 22 and 25 . 28 . 144 . 0 / 23

☒ **D** 25 . 28 . 148 . 0 / 22 and 25 . 28 . 146 . 0 / 23

Ans:

A, B, C & D

Available IP Address : 25.28.144.0/21

25.28.10010 ———— /21



Org. A

10010000.00000000/22

25.28.144.0/22

Org. B

10010100.00000000/23

25.28.148.0/23

ISP

10010110.00000000/23

25.28.150.0/23

25.28.10010 - - - . - - - - - /21

org. A

10010100.0000000000/22
25.28.148.0/22

org. B

10010000.0000000000/23
25.28.144.0/23

ISP

10010010.0000000000/23
25.28.146.0/23

[MCQ]



#Q. Which one of the following CIDR prefixes exactly represents the range of IP addresses "170.120.200.0" to "170.120.207.255"?

A 170.120.200.0 / 20

170.120.11001000.00000000
21 bit

☒ **B** 170.120.200.0 / 21

170.120.11001111.11111111
21 bit 11 bit

C 170.120.200.0 / 22

D 170.120.200.0 / 23

CIDR Prefix

170.120.200.0 / 21

Ans: B

#Q. Consider routing table of a IPv4 router shown below :

Destination	Mask	Interface
80 . 82 . 0 . 0	255 . 255 . 0 . 0	1
80 . 82 . 64 . 0	255 . 255 . 192 . 0	2
80 . 82 . 80 . 0	255 . 255 . 240 . 0	3
<u>80 . 82 . 84 . 0</u>	<u>255 . 255 . 252 . 0</u>	4 ✓
Default		5

Consider an IPv4 datagram contains destination IP address "80 . 82 . 84 . 86" arrive at the router. On which interface will it be forwarded ?

84 → 010100
252 → 1111100

Ans = 4

[NAT]



#Q. Consider routing table of a Classless Inter-domain Routing (CIDR) IPv4 router shown below :

210 → 11010010
192 → 11000000
192 ← 11000000

Destination	Interface
220.215.0.0 / 16	1
220.215.192.0 / 18	2 ✓
220.215.200.0 / 20	3 ✗
220.215.210.0 / 22	4 ✗
Default	5

210 → 11010010
252 → 11111100
208 ← 11010000

210 → 11010010
240 → 11110000
208 ← 11010000

Consider an IPv4 datagram contains destination IP address "220.215.210.205" arrive at the router. On which interface will it be forwarded ?

Ans = 2



THANK - YOU